

A CONVERSATION BASED APPROACH TO LEARNING SCOTTISH GAELIC

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Introduction

Although Scottish Gaelic (Gàidhlig) is an endangered language, there is a growing number of resources available to learn it. Many of these tools don't work for some people's learning styles. There is a lack of online accessible interactive tools that don't require a large time commitment. Examples are Duolingo, SpeakGaelic, and LearnGaelic. Some of these tools lack in-context material, interactivity, or retention mechanics.

Objective

- Identify the theory surrounding language learning motivation, mobile language learning frameworks, retention and memory enhancement (MALL), and language learning techniques.
- Design and implement a language learning application. Contextual, interactive, and memory effectiveness.
- Evaluate effectiveness in terms of user experience and MALL framework compatibility (PF4M [1]).

Methodology

- 1 The site utilised pre-scripted conversations per unit which the user then completed several activities based on the conversation content.
- 2 The topics of each unit were selected based on common progression in resources like LearnGaelic and SpeakGaelic, while being influenced by most common word databases.
- 3 The spaced repetition algorithm Free Spaced Repetition Scheduler (FSRS) was employed to help with memory retention.

Conversations

Feasgar math!

Good afternoon/evening!

Feasgar math. Ciamar a tha thu?

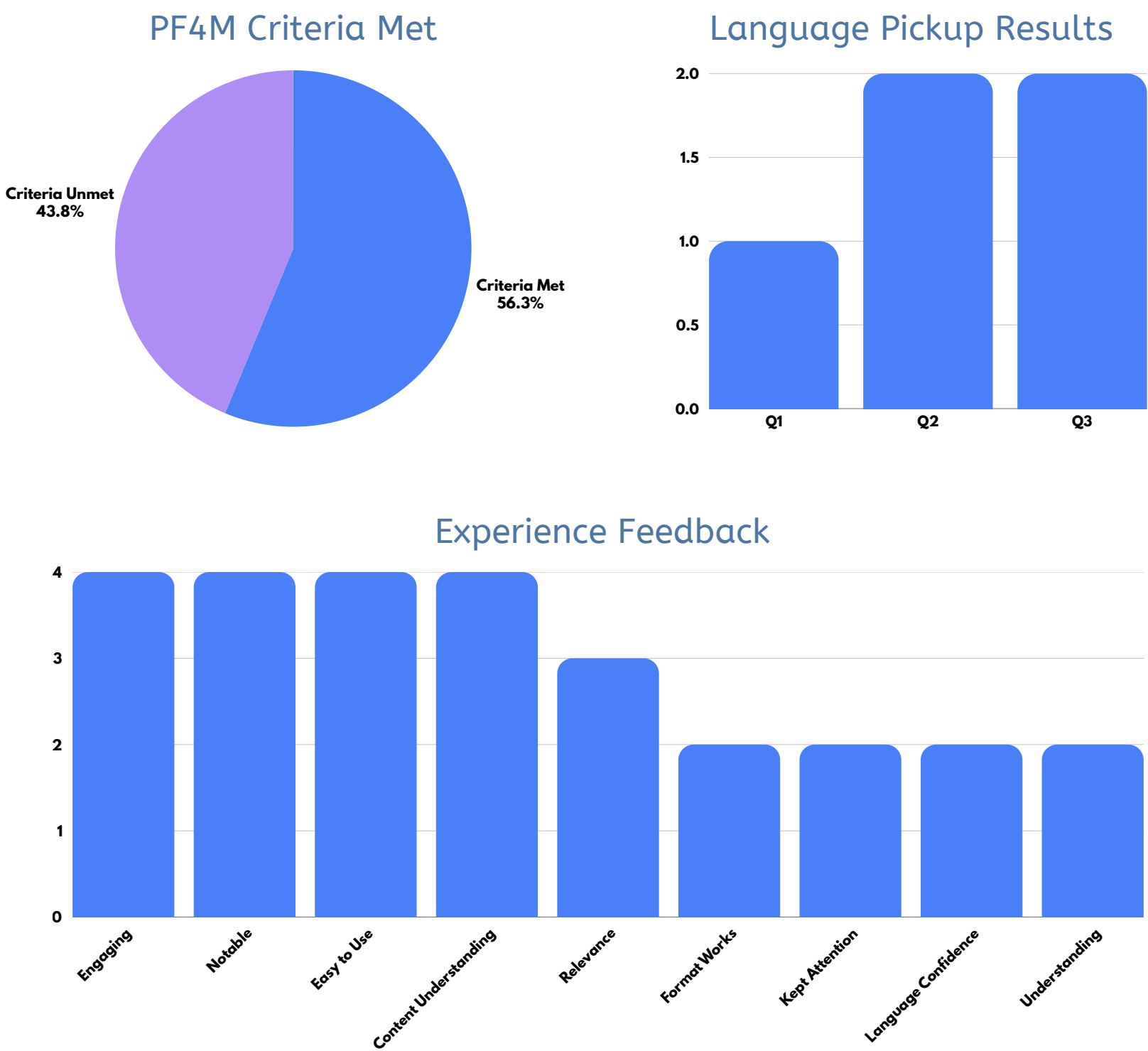
Good afternoon/evening. How are you?

Tha mi gu math, tapadh leat. Ciamar a tha thu?

I am good, thank you. How are you?

Tha mi gu math cuideachd, tapadh leat.

I am good too, thank you.



Results & Analysis

The results from the user survey and MALL framework compatibility were promising. 100% of users found the site engaging and enjoyed the format. While 50% of users thought that the format improved confidence, understanding, and attention. Users correctly answered the quantitative language quiz. The results suggest we need to refine the design to create a memorable experience rather than merely promote memory retention.

Conclusion

The project successfully implemented an interactive language learning experience that is proven to help people learn Gàidhlig language in an engaging and accessible way; however, in order for it to be a fuller, more motivating, mobile, and comprehensive experience, further design and research are required, and a larger cohort of users is needed to fully assess its performance.

Future Work

- Design more engaging content, progression, and quiz activities to help with more effective language learning.
- Incorporate more academic theory and improve compatibility with language-learning frameworks.
- Design and include other learning media than text, including video, audio, speech recognition, and LLMs.
- Include better personalisation on a unit-level so content can adapt better to individual users.

References

[1] X. Wang, A. B. Hamat, and N. L. Shi, 'Designing a pedagogical framework for mobile-assisted language learning', Heliyon, vol. 10, no. 7, p. e28102, Apr. 2024, doi: [10.1016/j.heliyon.2024.e28102](https://doi.org/10.1016/j.heliyon.2024.e28102).